

Ruthvika Mamidyala

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EDUCATION

Master of Science, University of North Carolina, Charlotte

Aug 2024 – Dec 2025 | Charlotte, USA

Computer Science

Relevant Coursework: Data Mining, Big Data, Computer Networks, DataBase Systems, Visual Analytics, Software Systems Design and Implementation.

PROFESSIONAL EXPERIENCE

Data Scientist, TenXengage Inc

Jul 2022 – Jul 2024 | Hyderabad, India

- Fine-tuned LLMs (GPT-4, LLaMA) using SFT, DPO, and RLHF to generate context-aware deal recommendations and partner insights, achieving 92%+ accuracy and 28% higher deal closure rates.
- Optimized models via distillation, quantization, and pruning, reducing inference latency by 40% while maintaining production-grade precision and throughput.
- Built and deployed PyTorch- and Hugging Face-based ML pipelines for partner profitability prediction, covering full lifecycle from data acquisition to productization.
- Architected microservices and backend pipelines using AWS SageMaker, Lambda, Kubernetes, and Docker, integrating Spark for distributed data processing.
- Implemented semantic retrieval and knowledge graph enrichment using LangChain, Pinecone, and Neo4j, powering low-latency, high-relevance partner intelligence queries.

Data Scientist, Geni Community Pvt Ltd

Oct 2021 – Jun 2022 | Hyderabad, India

- Engineered a data analytics pipeline integrating SQL, Python, and Power BI to process 1M+ patient records and extract key mortality-risk indicators from raw hospital data.
- Conducted exploratory and statistical analysis to identify comorbidity patterns and high-impact predictors, improving model interpretability and clinical insight quality by 30%.
- Built and validated XGBoost and Random Forest models achieving AUC 0.92 / F1 0.88, increasing early-risk detection accuracy by 25% compared to baseline.
- Automated ETL refresh and data synchronization between SQL and Power BI, ensuring continuous data availability with zero manual intervention.
- Delivered interactive Power BI dashboards for clinicians to visualize patient cohorts and risk trends, reducing decision latency by 30% and reporting time by 40%.

PROJECTS

GenAI Multi-Agent Knowledge Assistant (MCP, A2A, GPT-4o)

- Designed and deployed a multimodal RAG system that processes 1,000+ pages of user manuals using OCR, structured parsing, and GPT-4o for semantic correction — extracting 95%+ relevant data.
- Invented a context-linking layer that fuses visual elements (tables, figures) with section metadata, boosting semantic recall by 35%.
- Deployed a modular service cluster on Kubernetes with Docker containers, achieving zero downtime and 40% latency reduction.
- Integrated Pinecone-based vector retrieval for context-aware responses, sustaining 90%+ precision on hybrid visual-textual queries.

Google Talos, Frontend using React, Javascript, HTML, CSS

- Engineered dynamic, high-performance web pages using React.js, JavaScript (ES6+), and Redux, improving responsiveness and user engagement by 20% while cutting load times by 35%.
- Developed modular, reusable UI components and scalable state-management logic that accelerated feature releases by 3 weeks and reduced development complexity.
- Revamped a legacy front-end stack by migrating to a modern React architecture, improving maintainability, scalability, and reducing code redundancy by 40%.
- Integrated RESTful APIs with optimized client-side caching and asynchronous rendering, achieving a 25% improvement in page load performance and smoother data delivery.

Assessment Of Factors Impacting Mortality Risk, Python, Data Analytics, Excel, PowerBI

- Designed a risk assessment engine in Python with end-to-end data prep, model training, and visualization — increasing clinical decision precision by 25%.
- Built interactive Power BI dashboards for risk segmentation and outcome tracking, enabling clinicians to act on insights 30% faster.
- Integrated automated refresh pipelines between Excel and Power BI, ensuring continuous updates and zero manual intervention.

SKILLS

Programming and Data Science: Python, JavaScript, SQL, Pandas, NumPy, Scikit-Learn, TensorFlow, XGBoost, Random Forest, NLP

AI & Data Engineering: LangChain, FAISS, Pinecone, Vector Databases, RAG Pipelines, MongoDB, MySQL, Snowflake, BigQuery, ETL Pipelines, Data Modeling, multi-agent (MCP, A2A), RAG pipelines

Visualization and BI: Power BI, Tableau, Excel (VLOOKUP, Pivot Tables), Data Storytelling

Web & Application Development: React, HTML5, CSS3, Tailwind CSS, Bootstrap, Node.js, REST APIs.